

Case Study: Bill Walsh's 49ers and the West Coast Offense

How a team that couldn't run the ball made the short pass its running game, and how its coaches spread across the league.

Third-and-3 at the Dallas 6, fifty-eight seconds left in the 1981 NFC Championship Game, the 49ers behind 27–21. Joe Montana takes the snap and sprints to his right with three Cowboys chasing him toward the sideline. His first receiver has slipped. Montana pump-fakes, backpedals, and throws the ball high toward the back of the end zone, so high that the Cowboys' defensive end Ed "Too Tall" Jones, who had left his feet on the fake, can't reach it. Dwight Clark jumps and gets his fingertips to it. It's the most famous catch in the sport, and it gets called a miracle, a desperation heave, a throwaway that a receiver happened to catch. Clark always said it was none of those things: it was a play from the book, run on the practice field so many times that he knew where the ball would be.¹ Which was it? And how did a team that had gone 2–14 two years earlier end up in that spot at all? The answers are the same answer, and it's the subject of this chapter.

You'll be able to:

- Explain how Walsh's **horizontal passing game** works as a substitute for the run, and say what it costs: why a five-yard throw can do a handoff's job and when it can't.
- Explain timing, yards after the catch and ball placement as *design goals*, built into every route and every drop, and spot each one on a broadcast replay.
- Describe how Walsh scripted games and ran practice, and explain why a timing offense depends on rehearsal more than a run-first one does.
- Break down "The Catch" (the 1981 NFC Championship) step by step: the call, the design, what went wrong, and why the play still worked.
- Trace the Walsh **coaching tree** from his assistants to the head coaches of the 2025 and 2026 seasons, and say what a tree does and doesn't tell you.
- Run a "West Coast census" on a modern game: count the short horizontal throws on early downs, and compare them with the league's numbers.

¹"The Catch (American football)," Wikipedia: [https://en.wikipedia.org/wiki/The_Catch_\(American_football\)](https://en.wikipedia.org/wiki/The_Catch_(American_football)) (January 10, 1982, Candlestick Park; Dallas led 27–21 after Doug Cosbie's touchdown; the drive began at the 49ers' 11 with 4:54 left and three timeouts, 89 yards, including Lenvil Elliott runs and a 14-yard reverse to Freddie Solomon; third-and-3 at the Dallas 6 with 58 seconds left; the 49ers' second timeout before the snap; formation: split backs, Solomon in the right slot, Clark outside him; the call "Change Left Slot – Sprint Right Option"; Solomon the primary receiver, who had scored on the same play earlier in the game, slipped; Clark's route diagonally left across the end zone and back right along the end line; Too Tall Jones, Larry Bethea and D.D. Lewis chasing Montana; the pump fake; Everson Walls covering Clark, with two interceptions and a fumble recovery earlier; the touchdown with 51 seconds left; Wersching's extra point; Pillers's sack and White's fumble, recovered by Jim Stuckey; Clark's view that it was a practised backup route, not a throwaway); San Francisco 49ers, "Celebrating the Anniversary of The Catch": <https://www.49ers.com/news/celebrating-the-anniversary-of-the-catch>. Accounts differ on the number of plays: Wikipedia counts 14, while newspaper retrospectives (for example the *San Francisco Chronicle*, January 9, 2022: <https://www.pressreader.com/usa/san-francisco-chronicle-late-edition-sunday/20220109/281925956365556>) give 13, so this chapter gives no play count. Super Bowl XVI (January 24, 1982; 49ers 26, Bengals 21): see the Walsh note.

i You'll need

This is a Level 4 case study, and it builds on two chapters in particular. One line each:

- **1978 and the Passing Revolution:** two 1978 rules (the Mel Blount rule and open-hand pass blocking) freed receivers past five yards and gave quarterbacks time; Coryell answered by stretching the field deep, Walsh by stretching it sideways.
- **Dropback Concepts:** a pass concept is a set of routes built to put one defender in a bind, read in a numbered order. You'll see drive (a shallow cross under a dig) and the curl-flat triangle again here.

Terms from elsewhere you'll meet often: the West Coast offense (WCO) and its long play calls, from Offensive Systems; slant, flat route, timing route, progression, YAC (yards after the catch) and the horizontal stretch, from Passing Fundamentals; slant-flat and the rhythm throw, from The Quick Game; the pro set, also called split backs, from Classic Formations; check-release, from Pass Protection; Cover 3, from The Cover 3 Family; the opening script, from The Game Plan; and EPA and success rate, from Downs, Distance, and Field Position.

Reading the diagrams: offense is blue circles (Q quarterback; H the halfback and F the fullback in split backs; X, Y, Z the receivers, Y the tight end; T G C G T the line). Defense is outlined squares (E end, T tackle, S M W the Sam, Mike and Will linebackers, CB corners, FS and SS safeties). Solid arrows are routes and runs, a thick dark arrow is the ball carrier, dashed orange arrows are defenders' drops, a bar at the end of a line is a block, and the dashed brown line is the pass. A green ring marks the player to watch; numbered green circles are the quarterback's progression. The left-edge numbers give depth in yards past the line of scrimmage. Plays are illustrative, typical of the era, not traced from film, unless a caption says otherwise.

The problem: 2–14, and no way to run the ball

In January 1979 the San Francisco 49ers hired a 47-year-old Stanford coach to fix a franchise that had just gone 2–14. Bill Walsh had waited a long time for the job. He had coached in pro football since 1966, with the Raiders (then an AFL team) and then for eight seasons with Paul Brown's Cincinnati Bengals, where he ran the passing game. When Brown retired from coaching after the 1975 season he picked another assistant, Bill "Tiger" Johnson, as his successor; Walsh resigned, and later said Brown had worked against him when other teams asked about him for head-coaching jobs. He spent 1976 as the Chargers' offensive coordinator, coaching a young Dan Fouts, and two seasons winning at Stanford before the 49ers called. He was also the general manager, which mattered: the man drawing the plays was the man choosing the players.²

What he inherited was a team with no good answer to the oldest question in football: how do you keep the ball? In the 1970s the answer was the run. A team that could gain four yards on first down and four more on second down controlled the clock, the field and the game, and passing was what you did when that failed. Walsh's 49ers couldn't do it. They didn't have the linemen to push other teams around, and Walsh had been in exactly this spot before. Of the young expansion Bengals he had coached in the late 1960s, he said: "We couldn't control the football with the run; teams were just too strong. So it had to be the forward pass, and obviously it had to be a high-percentage, short, controlled passing game."³

That sentence is the whole chapter in miniature. Walsh didn't decide to *pass more* instead of running. He decided that the pass could *be* his running game, doing the same job (four yards at a time, the clock moving, the ball kept) with players and plays he could actually win with. In Cincinnati he built a short, timed passing game for Virgil Carter, a quarterback without a deep arm, and in 1971 Carter led the league in completion percentage (the story is in Passing Fundamentals and The Quick Game). San Francisco gave him the chance to build a whole team around the idea, and the 1978 rules (the previous history chapter) had just made it far easier.

The first two seasons were rough: 2–14 in 1979 and 6–10 in 1980. In the 1979 draft Walsh took a Notre Dame quarterback, Joe Montana, in the third round, and a Clemson receiver, Dwight Clark, in the tenth. Two years later the 49ers went 13–3 and won

²"Bill Walsh (American football coach)," Wikipedia: [https://en.wikipedia.org/wiki/Bill_Walsh_\(American_football_coach\)](https://en.wikipedia.org/wiki/Bill_Walsh_(American_football_coach)) (born November 30, 1931; Raiders assistant 1966; Bengals assistant 1968–1975, resigning after Paul Brown named Bill "Tiger" Johnson his successor, and his claim that Brown "worked against my candidacy"; Chargers offensive coordinator 1976; Stanford 1977–78; 49ers head coach and general manager from 1979; season records 2–14 (1979), 6–10 (1980), 13–3 (1981), 15–1 (1984), 10–6 (1988); totals 92–59–1 in the regular season and 10–4 in the playoffs; Super Bowl XVI, 26–21 over Cincinnati, and XXIII, 20–16 over Cincinnati). Pro Football Hall of Fame, "Bill Walsh" (Class of 1993): <https://www.profootballhof.com/players/bill-walsh/>. The 49ers' 1978 record (2–14): "1978 San Francisco 49ers season," Wikipedia: https://en.wikipedia.org/wiki/1978_San_Francisco_49ers_season

³Walsh to Tom FitzGerald of the *San Francisco Chronicle*, quoted in "Walsh, Bill," *Encyclopedia.com*: <https://www.encyclopedia.com/people/sports-and-games/sports-biographies/bill-walsh> (the profile places the remark in its account of Walsh's Bengals offense, built for the expansion team he joined in 1968; the bracketed "[opposing]" in that source's version is dropped here).

The scheme answer: the pass as a run

Here is the idea in one term. A **horizontal passing game** is an offense built on short, timed throws toward the sidelines and across the field, caught a few yards past the line of scrimmage and turned into gains by the run after the catch. Its throws are so short that they're often described as extended handoffs.⁵ The name contrasts it with the *vertical* passing of Don Coryell's Chargers, which attacked down the field.

What does a run actually do for an offense? Three things. It gains a few yards with little risk of losing many. It puts the ball in a fast player's hands while he's moving forward, so he can make a defender miss. And it attacks the defense at a spot the offense chose. A short pass to the flat does all three, and Figure 1 shows the comparison that Walsh's offense made every week: two ways to get the halfback to the right edge with the ball.

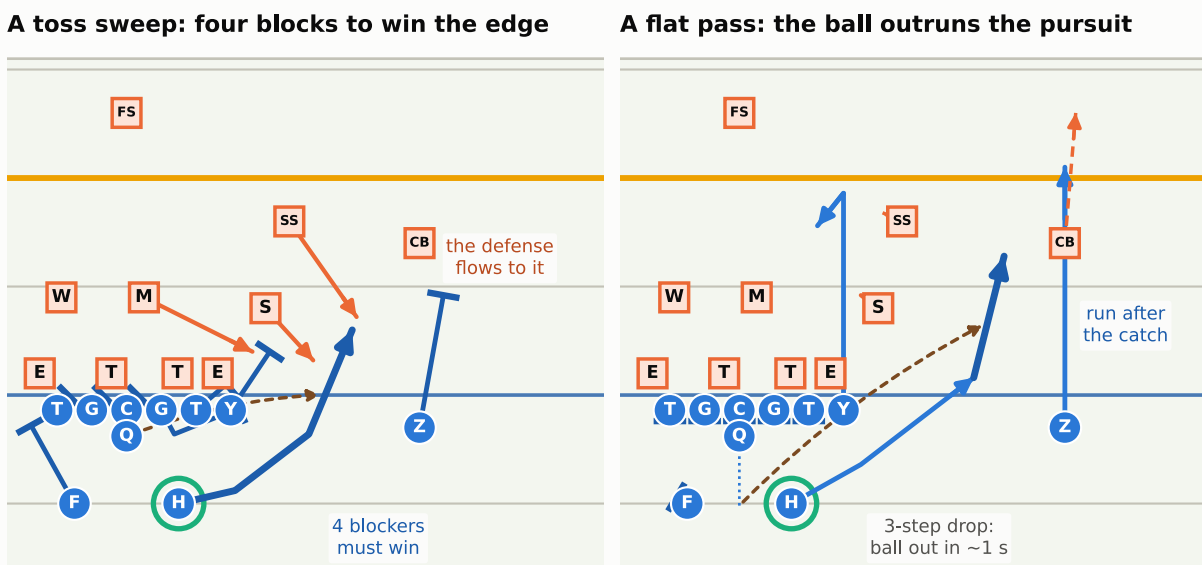


Figure 1: The same job, two ways (illustrative, 1980s 4-3, Cover 3). Left: a toss sweep. The halfback (H, ringed) needs the right tackle and tight end to reach their men, the right guard to pull around them and the flanker (Z) to block the corner back, and he gets to the edge just as the Mike, the Sam and the strong safety, flowing with him, arrive; one missed block and he's tackled for a loss. Right: the halfback runs a flat route instead and the quarterback throws it on a three-step drop. The ball beats the flow: it arrives about a second and a half after the snap, near the sideline, where the only nearby defender (the strong safety) has dropped back under the tight end's curl. No one has to win a block at the edge. That is the horizontal passing game: a pass that does a run's job.

Look at what the pass *removes*. The toss sweep is a bet that four blockers each win their fight in the same second; the flat pass is a bet that one throw is accurate. Against a defense that's bigger and stronger than your line, the second bet is the better one, and in 1979 the 49ers' line was the weak point. The pass also changes *who* does the tackling. On a run the first defenders to arrive are linemen and linebackers, the biggest men on the field. On a quick throw to the edge the tackler is usually a defensive back, 30 or 40 pounds lighter, and often backing up when the ball arrives.

Then there is the horizontal stretch, which is what makes the throw hard to take away. A zone defender who covers the flat can't also cover a slant breaking inside him, so the offense puts a receiver on each side of him and throws to whichever he leaves. Spread that across the field (the backs releasing to both flats, receivers crossing underneath) and every short defender has two receivers to worry about. The defense can't add enough players underneath to cover them all without emptying the deep part of the field. That, Walsh argued, was when the long throw came open.

⁴"1979 NFL draft," Wikipedia: https://en.wikipedia.org/wiki/1979_NFL_draft (Joe Montana, Notre Dame, third round, 82nd overall); "Dwight Clark," Wikipedia: https://en.wikipedia.org/wiki/Dwight_Clark (Clemson, tenth round, 249th overall; 6 ft 4 in).

⁵The same *Encyclopedia.com* profile describes his Bengals offense, the seed of the West Coast offense, as "passes so short they were effectively extended handoffs." See the previous note.

! Why it exists, and what it costs

The horizontal passing game exists because **a team that can't win at the line of scrimmage can still win in space**. It trades the run's physical contest for a contest of precision and speed: the ball gets to the edge before the defense does, and the tackle is made, or missed, by a smaller man in open field.

The bill comes in four parts. **It needs an accurate quarterback**, more than a strong-armed one: a throw a yard behind the receiver turns a five-yard gain into a tackle at the catch. **It needs receivers who run after the catch**, because the throw itself only travels a few yards. **A short throw can still be intercepted**, and a linebacker who reads it and jumps it has a clear run the other way. And **it gets squeezed**: once a defense stops worrying about the deep ball, its underneath defenders can crowd the short throws, which is why Walsh insisted the offense keep threatening deep. A West Coast team that only throws short isn't running Walsh's offense.

🔥 Common misconception

"The West Coast offense is dink-and-dunk." Walsh hated the word *finesse*, which writers attached to his offense, and the record supports him.⁶ The short throws were the setup, not the point: they pulled linebackers and safeties forward, and the 49ers then threw over them. Jerry Rice caught 22 touchdown passes in twelve games in 1987, many of them long runs after slants and deep throws, which is not what dink-and-dunk looks like.⁷ What is true is that the offense was built so that a short gain was an acceptable result, which most 1970s passing games couldn't say.

Design goal 1: timing

Walsh's offense is a timing offense, the purest version of the idea you met in Passing Fundamentals: the quarterback's drop and the receiver's route are designed as one unit, so that the ball leaves his hand at the moment the receiver breaks, and arrives before the defender can react. That chapter's chart of drops and route depths is the West Coast offense's skeleton: a three-step drop for routes that break at 5 or 6 yards, a five-step drop with a hitch for routes at 10 to 14, a seven-step drop for the deep comebacks and posts.

Two things made Walsh's version different from a generic timing game.

The quarterback's feet were his reading clock. Each step of the drop meant a receiver in the progression. On the last step of a quick drop he threw to the first receiver if he was open. If not, the hitch step that followed was his permission to go to the second, and so on down to the checkdown. This is why the offense could be taught: a quarterback didn't have to *decide* when to give up on a receiver, because his feet decided for him.

The backs were receivers, not spare blockers. The original West Coast offense lined up in split backs: one back behind each tackle, a tight end, and two wide receivers.⁸ On a pass play both backs had rules that turned them into receivers. Each checked his blocking assignment first (a linebacker who might blitz) and released into a short route if nobody came, which is the check-release. A defense that covered all five possible receivers had to account for both backs, and a defense that blitzed found a back already in its way. Figure 2 shows one such play: the slant-flat, run from split backs, with the halfback as the flat receiver.

⁶"West Coast offense," Wikipedia, cited above: Walsh resented the label, "especially the word 'finesse,'" and called *Sports Illustrated's* Paul Zimmerman to protest that "that wasn't his offense."

⁷Pro Football Hall of Fame, "Jerry Rice" (Class of 2010; 22 touchdown catches in the strike-shortened 12-game 1987 season, then a record): <https://www.profootballhof.com/players/jerry-rice/>

⁸"West Coast offense," Wikipedia: https://en.wikipedia.org/wiki/West_Coast_offense ("originally the offense used two split backs," with five players on one side of the formation and four on the other; short, high-percentage passing; the quarterback's three reads before checking down to a back or tight end).

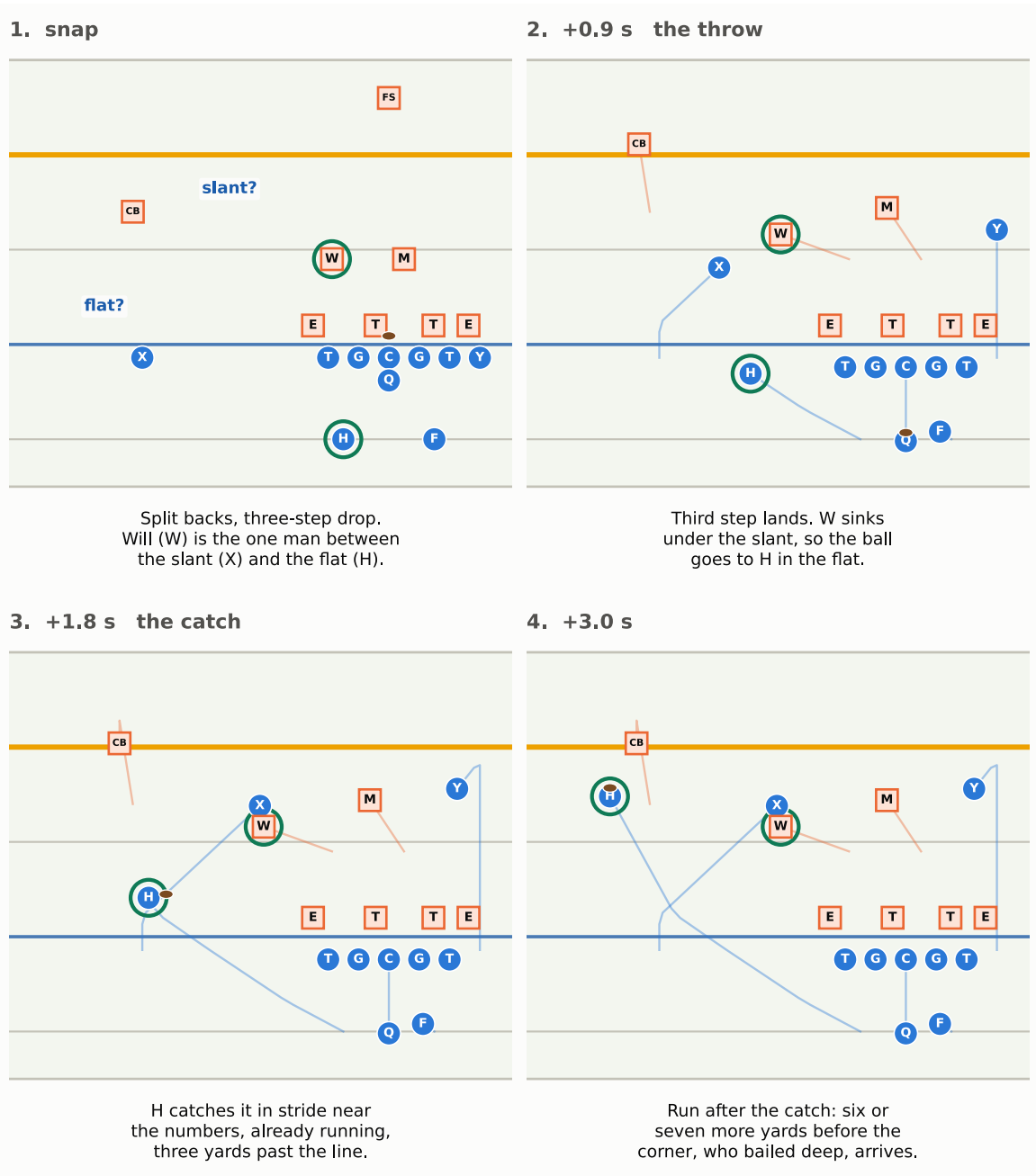


Figure 2: Slant-flat from split backs, with the halfback in the flat (simulated, BDB-format tracking; illustrative 1980s 4-3, Cover 3). The quarterback takes three steps from under center. X runs a slant; the halfback (H, ringed) releases from the backfield to the flat. The Will (W, ringed) is the one defender between them, and here he sinks inside under the slant, so the throw goes to H. H catches it on the run near the numbers, about three yards past the line, a second and three-quarters after the snap, and turns up the sideline for six or seven more yards before the cornerback, who had backed up to guard his deep third, can come back down.

Faint lines are where each player has been; the brown dot is the ball.

Notice when the ball leaves: on the third step, before the halfback has even reached the flat. The quarterback isn't waiting to see the halfback open; he's reading one defender, the Will, and throwing to a spot the halfback will reach. That's what *timing* buys. The defense gets no time to react to the throw, because the throw is already in the air when the receiver arrives. And it's what timing costs: if the halfback is a step late because a linebacker bumped him, or the quarterback's third step lands a beat slow, the whole thing falls apart. (The same concept from a modern shotgun spread, with a slot receiver instead of a back in the flat, is in The Quick Game.)

The quick game was only the front door, though. The heart of the offense was the five-step drop thrown on time to receivers crossing at 8 to 15 yards, with a back underneath as the safety valve. The classic West Coast version of drive in Figure 3 shows the

shape: a shallow cross under a dig from the same side, a high-low read of the middle linebacker, and both backs releasing to the flats as the third and fourth reads.

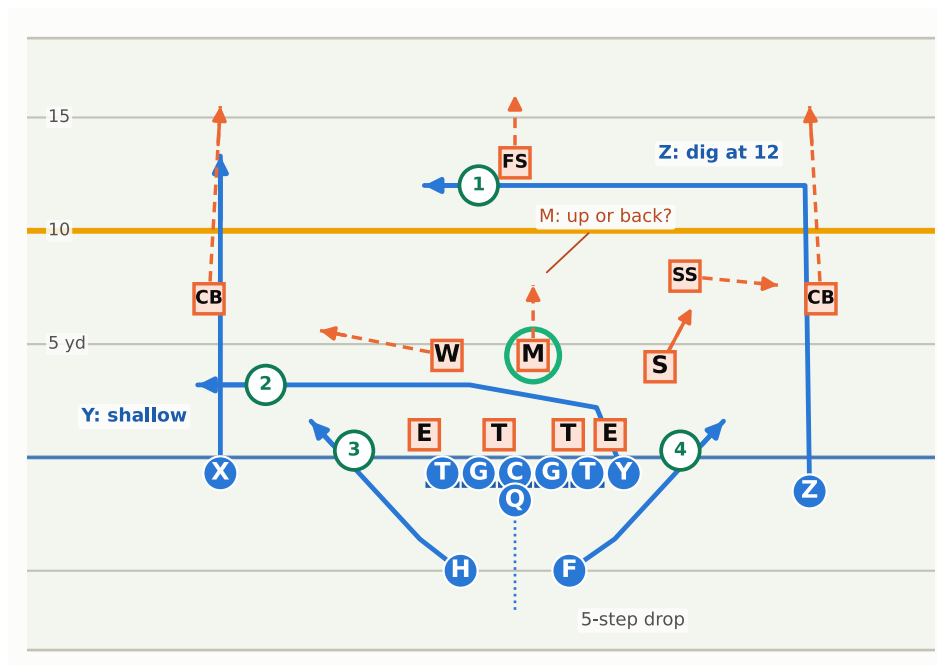


Figure 3: Drive from split backs, West Coast style (illustrative, 1980s 4-3, Cover 3). The tight end (Y) runs a shallow cross at about 3 yards; the flanker (Z) runs a 12-yard dig behind him on the same line. The Mike (M, ringed) can come up for the shallow or drop under the dig, not both, and the quarterback reads him on a five-step drop: (1) the dig if the Mike comes up, (2) the shallow if he drops. Both backs check for a blitz, then release to the flats as (3) and (4), a checkdown on each side. X's go route takes the left cornerback deep, and the free safety has the deep middle. Every receiver is running when he catches the ball.

Compare it with the plate in Dropback Concepts, drawn from a modern spread. The concept is the same; what's different is where the outlets come from. In the spread they're receivers; in Walsh's version they're the two backs, who were also the offense's protection a second earlier. That double duty is the West Coast offense in one picture: the same eleven players doing two jobs, so the defense has to cover more men than it expected.

Design goal 2: yards after the catch

Every one of these throws asks the receiver to keep running after he catches it. That's not a happy accident; it's what the routes were built for. The slant catches the receiver moving across the field at full speed. The shallow cross catches him running away from the defender who started across from him. The flat route catches the back moving toward the sideline with the cornerback backpedaling away from him. In each case the receiver is already in motion when the ball arrives, and the nearest defenders are either behind him or moving the wrong way.

A run after the catch is worth more than it looks. On the slant in particular, the receiver catches the ball in the gap between the underneath zone and the deep zone, the only defenders who can tackle him are a step behind, and one missed tackle can turn seven yards into seventy. That's how the 49ers' short offense produced long touchdowns.

It also explains the 49ers' roster. Walsh's best players were the ones who could turn a short catch into a long gain:

- **Jerry Rice**, a first-round pick in 1985 out of Mississippi Valley State, who finished his career with more catches, receiving yards and touchdowns than anyone in NFL history (1,549, 22,895 and 208 total touchdowns).⁹ His signature was the slant, caught in stride and taken the distance.
- **Roger Craig**, a second-round pick in 1983, a running back who was also one of the best receivers on the team. In 1985 he became the first player to run for 1,000 yards and catch passes for 1,000 more in the same season, and he led the league with 92 receptions.¹⁰ A running back leading the NFL in catches is the West Coast offense's fingerprint in a single statistic.
- **Dwight Clark** and **Freddie Solomon**, who did most of the catching in the early 1980s, and later **John Taylor**, Rice's partner on the other side.

⁹Career totals (1,549 receptions, 22,895 receiving yards, 208 total touchdowns, all NFL records) as listed in "West Coast offense," Wikipedia, cited above; drafted in the first round, 16th overall, in 1985, from Mississippi Valley State: "1985 NFL draft," Wikipedia: https://en.wikipedia.org/wiki/1985_NFL_draft

The yards after the catch also mean the offense isn't at the mercy of how far the quarterback can throw. Montana, a third-round pick, wasn't known for arm strength; nor, before him, was Virgil Carter. What both could do was put the ball in the right place at the right time, which leads to the third design goal. (YAC as a statistic, and the modern version that measures it against what a typical receiver would have gained, *YAC over expected*, belong to the Shanahan case study, which is where the 49ers' current offense picks this thread back up.)

Design goal 3: ball placement

If the receiver has to keep running, *where* the ball arrives matters as much as whether it arrives. A pass thrown to the receiver's back hip makes him stop, turn and reach behind him, and by the time he's turned upfield again the defender is on him. The same pass thrown a foot in front of him, at chest height on his upfield shoulder, lets him catch it without breaking stride. Walsh coached quarterbacks to throw to a *spot*, placed for the run after the catch and away from the nearest defender, and receivers to finish their routes so the spot would be right. Figure 4 shows the three versions of one slant.

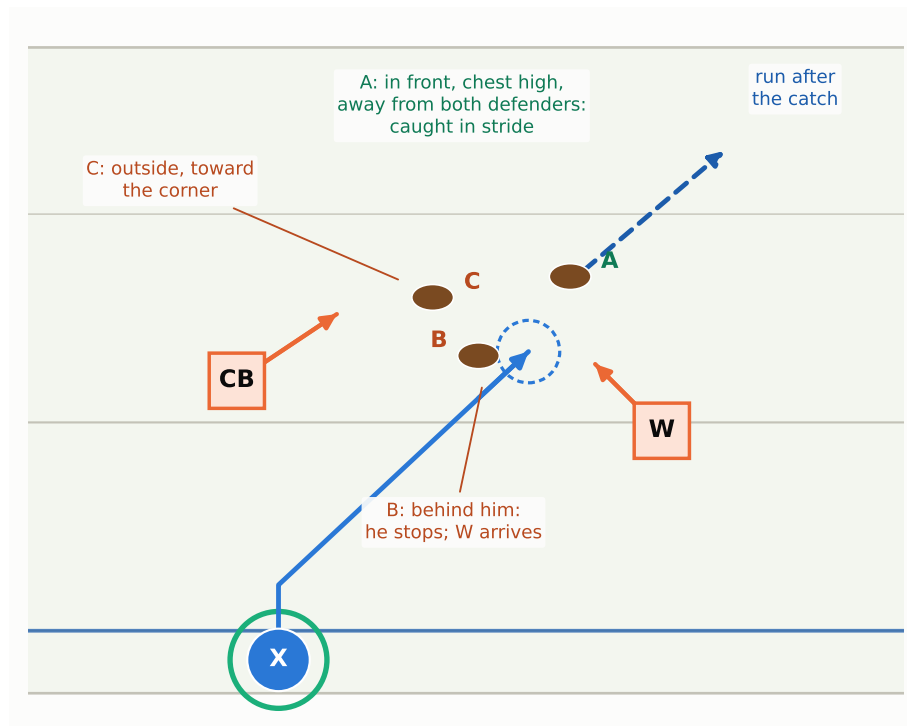


Figure 4: Ball placement on a slant (illustrative close-up of the offense's left side). X (ringed) breaks inside at about 2 yards and runs across the field. The cornerback trails behind and outside him after bailing; the Will linebacker is inside, closing. A: the right throw, a step in front of X at chest height, on the side away from both defenders. He catches it running and the dashed arrow is his run after the catch. B: a throw behind him. He has to stop and reach back, and the Will arrives as he turns. C: a throw outside him, toward the cornerback, who can drive through X's arm or take the ball.

The difference between A and B is perhaps a yard and a quarter of a second, and it is the difference between the play Walsh designed and a two-yard gain. That's why the West Coast offense has a reputation for chewing up quarterbacks who aren't accurate: the plays are easy to draw and hard to execute, and everything depends on a throw that has to be right every time. It's also why its coaches have so often preferred an accurate quarterback with an average arm to the reverse. The cost of that preference shows up against defenses that crowd the short area and dare the quarterback to beat them deep.

¹⁰"Roger Craig (American football)," Wikipedia: [https://en.wikipedia.org/wiki/Roger_Craig_\(American_football\)](https://en.wikipedia.org/wiki/Roger_Craig_(American_football)) (second round, 49th overall, in the 1983 draft from Nebraska; 1,050 rushing and 1,016 receiving yards in 1985, "the first in NFL history to run and receive for at least 1,000 yards in the same season," with a league-leading 92 receptions).

i Madden vs. real life

In a video game, a pass is complete or it isn't; the receiver catches it and runs. In real life, a "completion" covers everything from a catch in stride with open grass ahead to a lunge at a ball thrown behind a receiver who is tackled where he lands. Box scores count them the same, and coaches don't. When you watch replays, look past *caught or not* to *where on the receiver's body the ball arrived*, and whether he had to slow down to catch it.

Scripting and practice: taking the guesswork out

Walsh is remembered for one planning habit above all: before every game he scripted his first fifteen or so offensive plays (sometimes as many as 25) and called them in order.¹¹ How a modern staff builds and uses an opening script, and whether scripted plays actually gain more, is the subject of *The Game Plan*; here the question is *why it mattered so much to this offense in particular*.

A timing offense can't improvise its way out of trouble. A run-first team that has a bad week of practice can still line up and run its best play harder. A West Coast team whose quarterback and receivers aren't in sync throws incompletions and interceptions. So Walsh treated the week as rehearsal for decisions, not just for plays. The script moved the play-calling decisions from the noise of the first quarter to the calm of the week, and it let the players rehearse the exact plays they'd run first, in order, until they ran them without thinking. It also turned the opening drives into a set of questions (how does this defense line up against split backs? what does it do when the flanker goes in motion?) whose answers set up the rest of the game.

The practices were built the same way. A 1990 review of Walsh's book on building the 49ers describes a plan in which he counted out the days of training camp and the season in advance, used four practice formats (from a hard two-and-a-half-hour session to a light half-hour the day before a game), and never scheduled two hard sessions in a row. Individual technique was drilled to the step, down to how many steps the quarterback takes and when he looks for each receiver. Plays were repeated until they were automatic, and answers to special situations ("third down and four at the goal line") were prepared in advance rather than invented during a game. His stated goal was that players "take the field to learn something each session," not just to have their toughness tested. He also coached the coaches, which is the beginning of the tree at the end of this chapter.¹²

There's a direct line from that practice field to the play this chapter opened with. The 49ers didn't draw up Sprint Right Option in the dirt with fifty-eight seconds left. They had run it, including its second option, all season.

! Why it exists

Scripting and rehearsal exist because **a precise offense is only as good as its worst repetition**. Walsh's plays left no slack in the timing, so he took the slack out of everything else: the order of the calls, the week's schedule, the footwork. The trade-off is flexibility. A script is a plan made before the defense has shown its hand, and a coach who follows it too faithfully can call a play the defense is already set to stop, which is why every script has exits for third down, the red zone and bad field position.

Signature play: The Catch (1981 NFC Championship)

January 10, 1982, Candlestick Park. The winner would go to Super Bowl XVI. The Dallas Cowboys had been to five Super Bowls in the previous eleven seasons; the 49ers hadn't been to one. Dallas cornerback Everson Walls had intercepted Montana twice, and his fumble recovery late in the game set up the touchdown that put Dallas ahead 27–21.

San Francisco got the ball at its own 11 with 4:54 left and all three timeouts. What happened next is a case for the horizontal passing game as a running game, played backwards. The Cowboys, expecting the 49ers to throw, played pass defense, and Walsh **ran the ball**: Lenvil Elliott, a backup running back, carried on several plays, and Freddie Solomon gained 14 yards on a reverse. In between came short throws to Solomon, Clark and the fullback Earl Cooper. A team famous for passing drove into field-goal range

¹¹The AP obituary (July 30, 2007) credits Walsh with popularizing "the practice of scripting the first 15 offensive plays of a game": <https://www.nfl.com/news/hall-of-fame-coach-bill-walsh-who-won-three-super-bowls-with-th-09000d5d800879e1>; "West Coast offense," Wikipedia, cited above, says he sometimes scripted up to 25.

¹²Richard J. Cattani, "Walsh on Football: The Corporate Game," *Christian Science Monitor* (November 8, 1990), a review of Walsh's *Building a Champion*: <https://www.csmonitor.com/1990/1108/ecatt08.html> ("everything was scripted"; "how many steps the quarterback takes"; "Plays were practiced until they became automatic"; options prepared for "third down and four at the goal line"; "Walsh counted out the days of the training camp and season"; "Four different practice formats," from "a grueling two-and-a-half-hour practice to a light, thirty-minute practice the day before a game," "never two hard sessions in a row"; Walsh: "It's vitally important that players take the field to learn something each session"; he "coached the coaches").

mostly by doing what the defense didn't expect. By the time they reached the Dallas 6 on third-and-3, with fifty-eight seconds left, the 49ers had used two timeouts.

The call was **Sprint Right Option**, the full call reported as "Change Left Slot – Sprint Right Option."¹³ The 49ers lined up in split backs with Solomon in the slot on the right and Clark wide outside him. The design:

1. **Montana sprints right.** A sprint-out moves the whole passing game toward one sideline. The quarterback gets away from the middle of the rush and runs toward his receivers, and the throws get shorter and simpler, because he only has to read half the field.
2. **Solomon is the first option,** on a short out toward the right corner of the end zone. The 49ers had scored on this exact play, to Solomon, earlier in the game.
3. **Clark is the second option.** He starts by crossing near Solomon's defender, a rub that makes it harder to stay with Solomon, then runs across the end zone to the left, stops, and comes back to the right along the end line, where the quarterback, still moving right, will find him.

Figure 5 recreates it.

¹³The full call is taken from the 49ers' own 1982 offensive playbook (p. 325), as cited in "The Catch (American football)," Wikipedia, cited above; the playbook is reproduced at Football XO's: <https://footballxos.com/download/offense/pro-offense/1982-San-Francisco-49ers-West-Coast-Offense-Bill-Walsh.pdf>. Clark's own retellings and the Associated Press obituary of Clark (Josh Dubow, June 4, 2018: <https://foxbaltimore.com/news/nation-world/former-49ers-wide-receiver-dwight-clark-dead-at-61>) use the short name, "Sprint Right Option," called by Walsh. The formation (split backs, Solomon in the right slot, Clark wide right) is from the Wikipedia article.

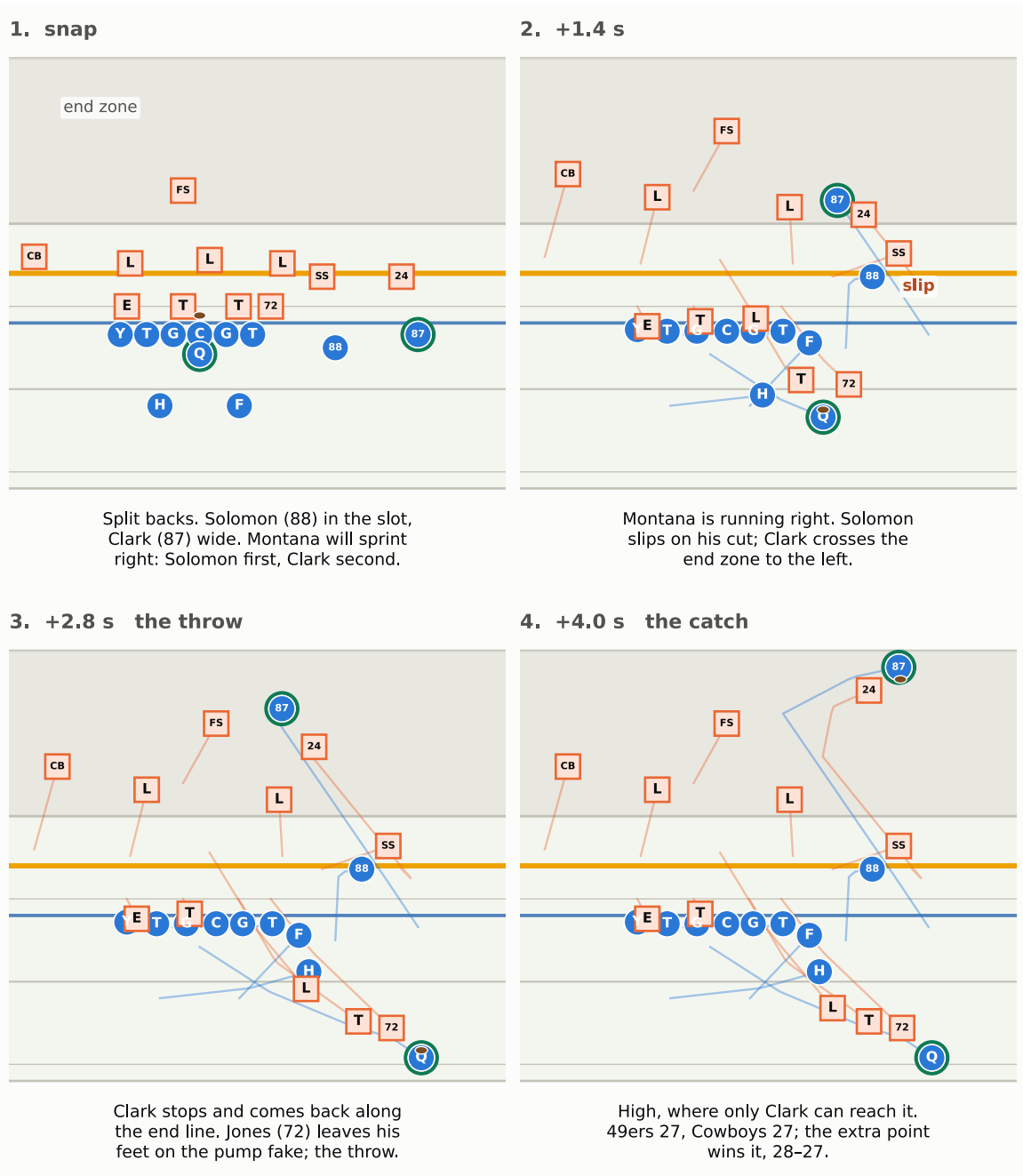


Figure 5: The Catch, recreated (simulated, BDB-format tracking; the 49ers' alignment follows contemporary accounts, while the Cowboys' alignment and the exact paths are illustrative, not traced from film). Third-and-3 at the Dallas 6, 58 seconds left. The 49ers are in split backs, with Freddie Solomon (88) in the right slot and Dwight Clark (87) wide right. Montana (Q, ringed) sprints right. Solomon, the first option, slips on his cut. Clark (ringed) crosses the end zone to the left, stops, and comes back to the right along the end line as Ed 'Too Tall' Jones (72) and two other Cowboys chase Montana toward the sideline. Montana pump-fakes, then throws high to the back of the end zone, where only Clark can reach it, over Everson Walls (24). The top edge of each panel is the end line.

Look at the third panel. That's the moment that gets called luck, and it's the moment the design is working. The first option is gone: Solomon has slipped, and the Cowboys have him covered anyway. Montana is out of room, running toward the sideline with three defenders closing. A quarterback without a plan throws the ball out of bounds or takes the sack, and on third down from the 6 with under a minute left, either one leaves the 49ers needing a touchdown on fourth down. Montana had a plan because the play had one. He knew Clark would be coming back along the end line, so he pump-faked to buy half a second, got Jones off his feet, and threw the ball where only a 6-foot-4 receiver jumping at the back of the end zone could catch it. If Clark couldn't, it would land out of bounds: no interception, and fourth down still to play.

Was it a throwaway that happened to be caught? People have argued about it ever since, and Montana's own description of the throw has varied over the years. Clark's answer never did: the route back along the end line was a practised backup, and he knew where the ball would come. Both things can be true. The *throw* was a quarterback under pressure putting the ball where nothing bad could happen; the *route* that put Clark under it was in the playbook and on the practice schedule. That is the practice-field argument of the last section in one snap.

The rest is history. Ray Wersching's extra point made it 28–27. Dallas still had 51 seconds and moved to midfield, but Lawrence Pillers sacked Danny White, White fumbled, and the 49ers recovered. Two weeks later they beat Cincinnati, the team Walsh had left, 26–21 in Super Bowl XVI.

! Why the sprint-out exists

The sprint-out (the quarterback running toward one sideline at the snap, with the routes flowing the same way) **trades half the field for simplicity and time**. The quarterback reads two or three receivers on one side instead of five across the field; he throws on the run, closer to his receivers; and he moves away from the rushers who started in the middle. Near the goal line, where the field is short and the end zone has edges a ball can be thrown over, those trades are worth more. The costs are real too: the whole back side of the field is gone, the throws are harder to make on the move, and a defense that knows the sprint is coming can chase it, as Dallas did. The modern cousins of the sprint-out are the bootleg and the moving-pocket throws you'll meet in Play-Action, Boots, and Screens.

The dynasty, and the system that outlived its coach

The Catch started a run that lasted longer than Walsh did. The table lists the 49ers' Super Bowl seasons of the next fourteen years.¹⁴

Season	Head coach	Regular season	Super Bowl
1981	Bill Walsh	13–3	XVI: beat Cincinnati 26–21
1984	Bill Walsh	15–1	XIX: beat Miami 38–16
1988	Bill Walsh	10–6	XXIII: beat Cincinnati 20–16
1989	George Seifert	14–2	XXIV: beat Denver 55–10
1994	George Seifert	13–3	XXIX: beat San Diego 49–26

Walsh retired after Super Bowl XXIII with a regular-season record of 92–59–1 and a 10–4 record in the playoffs. His defensive coordinator, George Seifert, took over and won two more titles with the same offense, the second with Steve Young at quarterback and Mike Shanahan as offensive coordinator. A scheme that keeps winning after its inventor leaves is a scheme that was taught, not just called.

i Film room: 49ers 1984 season, Super Bowl XIX

January 20, 1985, Stanford Stadium. The 15–1 49ers against Dan Marino's Dolphins, who had just set passing records. San Francisco won 38–16, Montana was the game's MVP, and Roger Craig scored three touchdowns, two of them on catches, the first player to score three in a Super Bowl.¹⁵

What to notice: the running back as a receiver. Watch Craig's routes out of the backfield: he isn't the last resort when everyone else is covered, he's a planned target, often lined up against a linebacker who can't run with him. A defense built to stop a two-back running team has the wrong players on the field to cover two backs who run routes.

¹⁴"Bill Walsh (American football coach)," Wikipedia, cited above (records for 1981, 1984 and 1988; Super Bowls XVI and XXIII); "Super Bowl XIX," Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_XIX (49ers 38, Dolphins 16); "Super Bowl XXIV," Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_XXIV (49ers 55, Broncos 10); "Super Bowl XXIX," Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_XXIX (49ers 49, Chargers 26; Seifert head coach, Mike Shanahan offensive coordinator); 1989 and 1994 regular-season records (14–2, 13–3): "George Seifert," Wikipedia: https://en.wikipedia.org/wiki/George_Seifert (49ers defensive coordinator 1983–1988, head coach 1989–1996).

¹⁵"Super Bowl XIX," Wikipedia, cited above (January 20, 1985, Stanford Stadium; Montana the MVP; Roger Craig's three touchdowns, two receiving, from 8 and 16 yards, and one rushing; the first player to score three touchdowns in a Super Bowl; Dan Marino's 1984 records of 5,084 passing yards and 48 touchdown passes).

January 28, 1990, New Orleans: 49ers 55, Broncos 10, still the most lopsided Super Bowl ever played. Montana threw five touchdown passes, then a Super Bowl record, three of them to Jerry Rice.¹⁶ It was Seifert's first season as head coach.

What to notice: the same offense without its inventor. The routes, the timing and the reads are Walsh's; the play-caller isn't. Watch how often the ball is out of Montana's hand before Denver's rush arrives, and how often Rice and Taylor catch it in stride and keep going.

The data fingerprint

The 1980s 49ers predate the play-by-play data this book uses (nflverse starts in 1999), so their fingerprint has to come from the season totals: an offense that completed passes at rates other teams couldn't match, threw relatively few interceptions for how often it threw, and had a running back lead the league in catches. Montana led the NFL in completion percentage five times (1980, 1981, 1985, 1987 and 1989), and his 1989 passer rating, 112.4, was a single-season record at the time.¹⁷

What *can* be measured is whether Walsh's central claim holds up today: that a short pass can do a run's job, and do it better. Figure 6 splits every early-down play in competitive games from 2023 to 2025 by what the offense did, a designed run or a pass of a given depth, and measures each by success rate (the share of plays that left the offense better off) and by EPA per play.

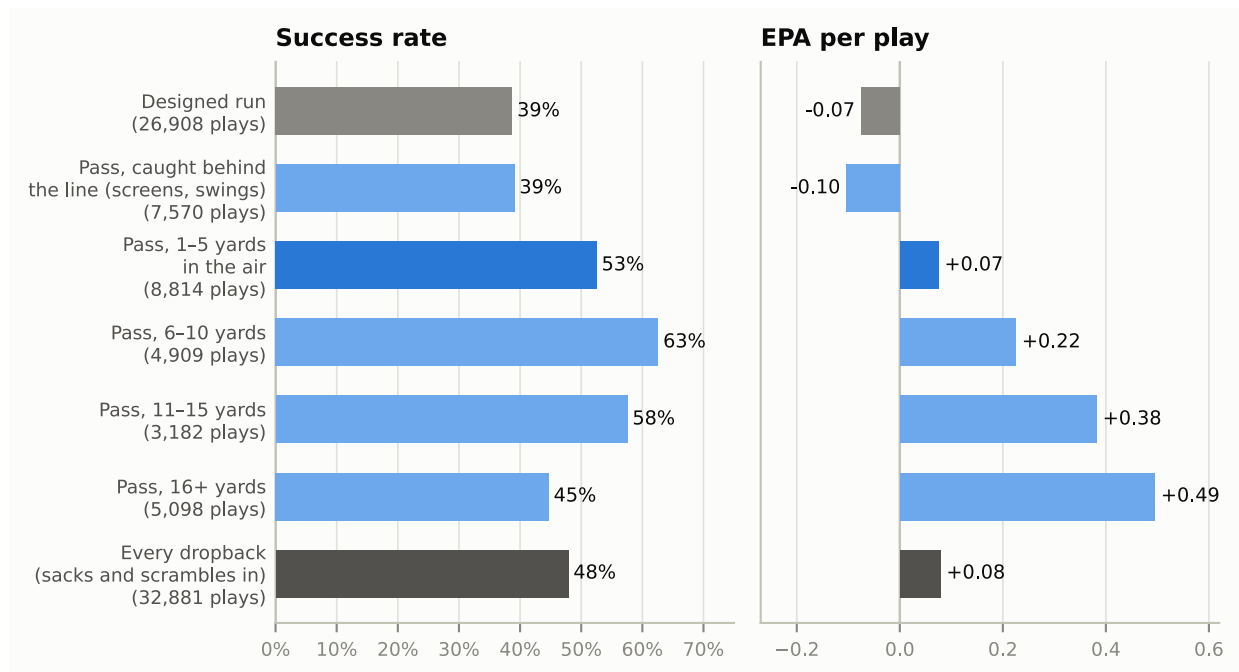


Figure 6: Walsh's claim, tested on modern data. Every first- and second-down snap with the win probability between 10% and 90%, 2023-2025 regular seasons. Left: success rate. Right: expected points added per play. A designed run (scrambles and kneels excluded) and a pass thrown behind the line (screens, swings) do almost exactly the same job. A pass caught one to five yards past the line, the heart of the horizontal passing game, succeeds far more often than either, and the six-to-fifteen-yard throws of Walsh's five-step game more often still. The last bar puts the sacks and scrambles back in, which is the fair comparison for 'passing' as a whole. Data: nflverse play-by-play via nflreadpy; course calculation.

Read the top three bars together. A designed run on an early down succeeded 39% of the time and gained 4.4 yards on average. A pass caught behind the line, a screen or a swing, succeeded 39% of the time: the same job, done about as well. A pass that travelled one to five yards in the air succeeded 53% of the time, with incompletions counted as failures. That's Walsh's argument in today's

¹⁶"Super Bowl XXIV," Wikipedia, cited above (January 28, 1990, Louisiana Superdome; Montana's five touchdown passes, then a Super Bowl record, three of them to Rice; the 45-point margin the largest in Super Bowl history, still the record through Super Bowl LX: see [planning/FACTS-current.md](#), §1, for the recent results).

¹⁷Pro Football Hall of Fame, "Joe Montana" (Class of 2000): <https://www.profootballhof.com/players/joe-montana/>; "Joe Montana," Wikipedia: https://en.wikipedia.org/wiki/Joe_Montana. The Wikipedia article lists his completion-percentage titles as 1980, 1981, 1985, 1987 and 1989, and calls his 112.4 rating in 1989 "the highest single-season passer rating in NFL history" at the time (Steve Young passed it in 1994). Pro Football Reference's season pages carry the same leaders but block automated fetches.

numbers: the short throw is a *better* run, more consistent and worth more expected points, and about 54% of the yards on those completions came after the catch.

Now read down. The best bar on the left belongs to throws of six to ten yards, which succeeded 63% of the time, and the 11-to-15-yard throws aren't far behind. That band is where Walsh's five-step game lived: the digs, crossers and curls you saw in Figure 3, thrown on time to a receiver on the move. The deep throws are worth the most per play but succeed less than half the time and carry most of the interceptions. And the last bar is the honest reminder that "passing" includes sacks and scrambles, which bring the average for every dropback back down: a short throw isn't better than *passing*, it's better than *running*, which is the claim Walsh made.

The league has absorbed the lesson so completely that it no longer looks like a system. In these three seasons 55% of early-down throws travelled five yards or less in the air, and 26% were caught at or behind the line. Every team in the league now runs some of Walsh's offense, which is the subject of the coaching tree two sections down.

Common misconception

"Short passes are just safe passes." The chart says something stronger and something weaker. Stronger: on early downs, short passes beat runs on consistency, which is what an offense needs to stay ahead of the chains. Weaker: a short pass is not automatically a *good* play. Its value comes from the defense leaving room underneath, and a defense that sits on short throws can make them fail. And on third-and-8 a five-yard catch is a punt, however efficient it looks on first down; there the deeper throw is the right one (Third Down).

How opponents responded

A system that wins three Super Bowls gets studied, and defenses found four answers. Each attacked one of the design goals.

Hit the receivers, and the quarterback. The timing depends on receivers reaching their spots on time, and inside five yards contact is still legal, so the first answer was to jam receivers at the line and knock them off schedule. The physical extreme was Bill Parcells's New York Giants, who beat the 49ers in the playoffs after the 1985 season (17–3) and the 1986 season (49–3), the second game remembered for the hit by nose tackle Jim Burt that knocked Montana out of it. One story credits Parcells with the name "West Coast offense" itself, as a jab after the first of those wins; another traces it to a 1993 remark by quarterback Bernie Kosar, publicized in *Sports Illustrated*, in which the label meant the Coryell family of offenses from San Diego and was then attached to Walsh's by mistake. Walsh disliked it either way.¹⁸¹⁹

Rush with safe pressure. A quick-throw offense beats an ordinary blitz by getting the ball out before it arrives, so defenses invented pressure that didn't leave a hole behind it. The Cincinnati Bengals, Walsh's Super Bowl opponent twice, made Dick LeBeau their defensive coordinator in 1984 and became the home of the zone blitz, which drops a lineman into the throwing lane of a hot route while a linebacker rushes, and which the NFL's own history describes as built to stop the West Coast timing game; the story is in Zone Blitzes.²⁰

Keep everything in front, and tackle. If the offense lives on yards after the catch, take away the "after." The Tampa 2 of the late 1990s and 2000s was built on fast linebackers and corners who sat short, let the short throw be completed, and swarmed the catcher. Its best-known coach, Tony Dungy, had been Dennis Green's defensive coordinator in Minnesota, and Green was a Walsh assistant: the tree was learning to stop itself (Cover 2 and Tampa 2).

Learn the language. The West Coast call is long and explicit, and once a few dozen coaches had learned it, they all spoke it. The best-known payoff came in Super Bowl XXXVII, after the 2002 season, when Jon Gruden's Tampa Bay Buccaneers beat the Oakland Raiders, the team he had coached the year before, 48–21. Rich Gannon threw five interceptions, and Raiders players said

¹⁸"West Coast offense," Wikipedia, cited above (Bill Parcells reportedly asking "What do you think of that West Coast offense now?" after the Giants' 17–3 playoff win over the 49ers in the 1985 season; Bernie Kosar's 1993 description of the Dallas Cowboys' offense as a "West Coast offense," meaning the Air Coryell system, publicized by Paul Zimmerman of *Sports Illustrated*, after which "a reporter mistakenly applied Kosar's statement" to Walsh's 49ers; Walsh's objection). The two stories are presented here as competing accounts; neither is settled. The course's Offensive Systems chapter gives the short version.

¹⁹"Giants–49ers rivalry," Wikipedia: https://en.wikipedia.org/wiki/Giants%E2%80%9349ers_rivalry (the Giants' 17–3 wild-card win after the 1985 season and 49–3 divisional win after the 1986 season, in which Jim Burt's hit knocked Montana out of the game with a concussion as he threw the pass that Lawrence Taylor returned for a touchdown; that game was played January 4, 1987).

²⁰"Dick LeBeau," Wikipedia: https://en.wikipedia.org/wiki/Dick_LeBeau (Bengals defensive coordinator 1984–1991); NFL.com, "NFL 100 Greatest Game Changers: Dick LeBeau/Zone Blitz": <https://nfl.com/100/originals/100-greatest/game-changers-60> (in 1984 the Bengals' new defensive coordinator began adapting their four-linebacker package "in an effort to stop the emerging West Coast offense"); "Zone blitz has come a long way since the advent of the 3-4," NFL.com: <https://nfl.com/news/zone-blitz-has-come-a-long-way-since-the-advent-of-the-3-4-09000d5d81e01d8e>.

afterwards that Oakland had kept much of Gruden’s terminology and that Tampa’s defenders seemed to know what was coming. (Derrick Brooks remembers it differently: Gruden played Gannon himself in practice, and “we didn’t know what play they was running, but we knew their personalities.”)²¹

Today’s version of the answer is the two-high safety shell of the 2020s, which takes away the deep throw first and rallies to everything short. The West Coast descendants answered that in turn, with under-center play-action and the outside-zone run game, which is where the Shanahan case study begins.

The coaching tree

A **coaching tree** is the family of coaches descended from one head coach, traced through who worked on whose staff. A branch runs from a head coach or coordinator to an assistant who later ran a staff of his own; because most coaches work for several bosses, one coach can sit on several branches. Walsh’s is usually called the largest in football. Figure 7 draws its offensive branches, and the two boxes underneath list the head coaches who came out of it in the 2025 and 2026 seasons.

²¹“Super Bowl XXXVII,” Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_XXXVII (January 26, 2003; Buccaneers 48, Raiders 21; Gruden had coached the Raiders through the 2001 season; Rich Gannon’s five interceptions, a Super Bowl record). The Raiders’ side: Josh Schrock, “Raiders claim Jon Gruden, Buccaneers knew their plays in Super Bowl XXXVII,” NBC Sports Bay Area, July 10, 2020: <https://www.nbcsportsbayarea.com/nfl/raiders-claim-jon-gruden-buccaneers-knew-their-plays-in-super-bowl-xxxvii/1325862> (Lincoln Kennedy: “Every level of the defense knew what we were doing”; Sam Adams recalling Buccaneers telling him they “cannot believe you’re using the same checks, and the same terminology”). The Buccaneers’ side: Josh Schrock, “How Gruden prepared Bucs to blow out Raiders in Super Bowl,” NBC Sports Bay Area, November 22, 2020: <https://www.nbcsportsbayarea.com/nfl/raiders/how-gruden-prepared-bucs-to-blow-out-raiders-in-super-bowl/1127303> (Derrick Brooks). Tim Brown has blamed Bill Callahan’s late change of game plan instead: ESPN, https://www.espn.com/nfl/story/_/id/8868196/tim-brown-bill-callahan-sabotaged-oakland-raiders-super-bowl-xxxvii-loss.

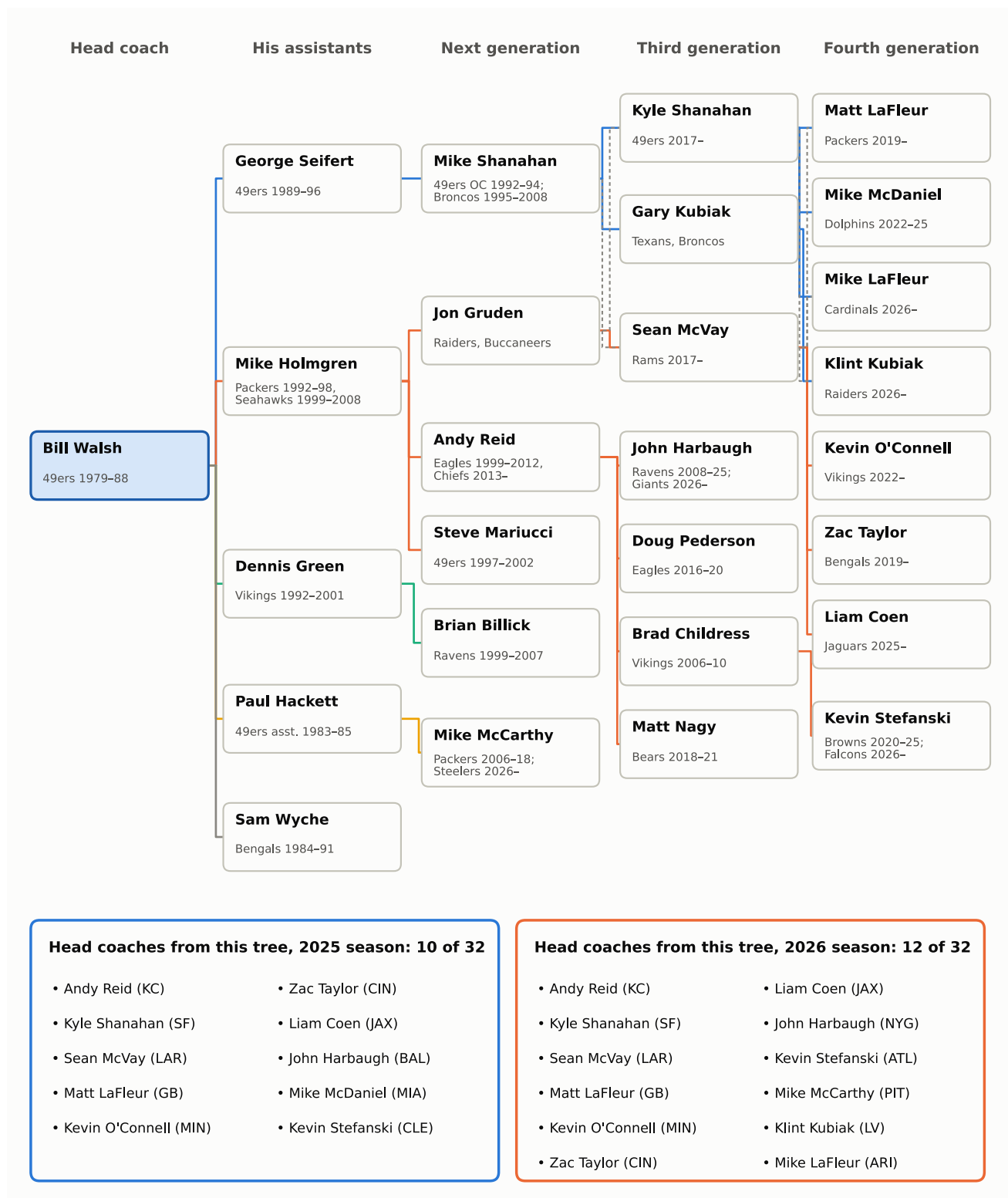


Figure 7: The Walsh coaching tree, offensive branches only (selected). A solid line means the coach on the right worked on the staff of the coach on the left, coloured by the branch it comes from (blue for Seifert and Shanahan, orange for Holmgren, green for Dennis Green, amber for Hackett); a grey dashed line marks a second important mentor. Years are head-coaching tenures unless marked. Mike Shanahan is the link to the modern wide-zone family: he never worked for Walsh, but was George Seifert's offensive coordinator from 1992 to 1994. The two boxes list the tree's NFL head coaches in the 2025 season and in the 2026 season, after a winter of moves (FACTS checked October 2026). Defensive coaches who came through these staffs (Mike Macdonald, Robert Saleh, DeMeco Ryans and others) are left off.

Read it from left to right and three big branches stand out.

Holmgren's branch is the one that kept the name. Mike Holmgren coached Walsh's quarterbacks from 1986 to 1988 and was Seifert's offensive coordinator, then won Super Bowl XXXI with Brett Favre's Packers (after the 1996 season). His Green Bay staffs produced Andy Reid, Jon Gruden and Steve Mariucci, and Reid's staffs in Philadelphia produced so many head coaches (John Harbaugh, Doug Pederson, Brad Childress, Matt Nagy, and others) that his branch is a tree of its own.²² It is also the branch that changed the most. Reid's Chiefs, covered in their own case study, still speak Walsh's language, but their offense has absorbed college spread concepts, run-pass options and a great deal of improvisation.

The Hackett branch is the quiet one. Paul Hackett coached for Walsh from 1983 to 1985 and later taught the system to a young quarterbacks coach, Mike McCarthy, at the University of Pittsburgh and then in Kansas City; McCarthy won Super Bowl XLV with the Packers and in 2026 is the head coach in Pittsburgh.²³

The Shanahan branch is the one that rules the league today, and it has the most interesting connection: Mike Shanahan never worked for Walsh. He was Seifert's offensive coordinator from 1992 to 1994, the years Steve Young ran Walsh's offense, then took it to Denver, married it to the outside-zone running game and bootlegs, and won two Super Bowls. One of them, Super Bowl XXXII after the 1997 season, came against Holmgren's Packers, 31–24: two branches of one tree meeting for the title.²⁴ Through his son Kyle, Gary Kubiak and Sean McVay (who also started as an assistant on Gruden's staff in Tampa), that branch produced most of the play-callers of the 2020s. That family is the subject of the Shanahan case study.

The 2025 and 2026 seasons, labelled separately. In the 2025 season, ten of the league's 32 head coaches came from these branches. The winter changed the list more than it shrank it. Mike McDaniels was fired in Miami and became the Chargers' offensive coordinator; John Harbaugh, fired in Baltimore after 18 seasons, became the Giants' head coach; Kevin Stefanski moved from Cleveland to Atlanta. Three more coaches from the tree became head coaches: Klint Kubiak, the 2025 Seattle offensive coordinator, in Las Vegas; Mike LaFleur, the 2025 Rams offensive coordinator, in Arizona; and Mike McCarthy in Pittsburgh. In 2026 the tree has twelve head coaches. Super Bowl LX, after the 2025 season, was won by a Seattle staff with both a defensive and an offensive root in the tree: head coach Mike Macdonald came up on Harbaugh's Baltimore staff, and offensive coordinator Klint Kubiak from the Shanahan family.²⁵

Common misconception

“Same tree, same offense.” A coaching tree records where coaches *learned*, not what they *run*. Reid's 2025 Chiefs, Shanahan's 49ers and Harbaugh's Ravens share a common ancestor and almost nothing on the field: shotgun spread and RPOs in Kansas City, under-center wide zone in San Francisco, and a quarterback-run offense in Baltimore. What travels down a tree is a way of working (the explicit language, the scripting, the attention to detail in practice) more than a playbook. Trees are also generous: almost every coach worked for several bosses, and the drawer chooses the line.

²²“West Coast offense,” Wikipedia, cited above (Holmgren as Walsh's quarterbacks coach 1986–1988 and Seifert's offensive coordinator 1989–1991, his Packers 1992–1998 and Super Bowl XXXI; Gruden, Reid, Harbaugh as an Eagles assistant 1999–2007, Pederson). Childress, Nagy and Mariucci's staff histories: “Brad Childress,” Wikipedia: https://en.wikipedia.org/wiki/Brad_Childress (Reid's Eagles quarterbacks coach 1999–2001 and offensive coordinator 2002–2005); “Matt Nagy,” Wikipedia; “Steve Mariucci,” Wikipedia: https://en.wikipedia.org/wiki/Steve_Mariucci (Holmgren's Packers quarterbacks coach 1992–1995).

²³“West Coast offense,” Wikipedia, cited above (Paul Hackett, 49ers assistant 1983–1985; taught the system to Mike McCarthy; McCarthy's Super Bowl XLV with the Packers); Green Bay Packers, “Mike McCarthy named Green Bay's 14th head coach” (2006), on his years under Hackett at Pittsburgh (1989–92) and as Kansas City's quarterbacks coach (1995–98): <https://www.packers.com/news/mike-mccarthy-named-green-bay-s-14th-head-coach-2452545> . McCarthy as Steelers head coach in 2026: course fact sheet (planning/FACTS-current.md, §9), from https://en.wikipedia.org/wiki/2026_NFL_season .

²⁴“Super Bowl XXXII,” Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_XXXII (January 25, 1998; Broncos 31, Packers 24; Mike Shanahan and Mike Holmgren head coaches). Mike Shanahan as the 49ers' offensive coordinator 1992–1994: “Mike Shanahan,” Wikipedia: https://en.wikipedia.org/wiki/Mike_Shanahan . McVay's first NFL job on Gruden's Tampa Bay staff (2008): “Sean McVay,” Wikipedia: https://en.wikipedia.org/wiki/Sean_McVay . Staff links in the tree figure: “Kyle Shanahan,” Wikipedia: https://en.wikipedia.org/wiki/Kyle_Shanahan (offensive quality control on Gruden's Tampa Bay staff 2004–2005; Houston under Gary Kubiak 2006–2009; Washington offensive coordinator under his father 2010–2013); “Kevin Stefanski,” Wikipedia: https://en.wikipedia.org/wiki/Kevin_Stefanski (joined the Vikings in 2006 “as an assistant to then-head coach Brad Childress”); “Liam Coen,” Wikipedia: https://en.wikipedia.org/wiki/Liam_Coen (McVay's Rams 2018–2020 and offensive coordinator 2022; Jaguars head coach from 2025).

²⁵Course fact sheet (planning/FACTS-current.md, §§1 and 9, checked October 2026), from https://en.wikipedia.org/wiki/2026_NFL_season (head coaches), <https://www.profootballrumors.com/2026/02/2026-nfl-offensive-defensive-coordinator-search-tracker> (coordinators), <https://kogo.iheart.com/content/2026-01-20-mike-mcdaniels-lands-nfl-coaching-gig-report> (McDaniels to the Chargers) and https://en.wikipedia.org/wiki/Super_Bowl_LX (Seattle's staff: Macdonald, Kubiak, Durde). Macdonald's years on Harbaugh's Baltimore staff: “Mike Macdonald (American football),” Wikipedia: [https://en.wikipedia.org/wiki/Mike_Macdonald_\(American_football\)](https://en.wikipedia.org/wiki/Mike_Macdonald_(American_football)) . Coen still the Jaguars' head coach in 2026: Jacksonville Jaguars, “Day after Week 1 win over Browns”: <https://www.jaguars.com/news/day-after-week-1-win-over-browns-coen-pleased-with-fast-week-1-start> ; Zac Taylor retained by Cincinnati for 2026: FOX19, “Zac Taylor, Duke Tobin will be back for 2026 season” (January 5, 2026): <https://www.fox19.com/2026/01/05/zac-taylor-duke-tobin-will-be-back-2026-season/> . The 2025 list counts head coaches who started the 2025 season.

There's one more branch the figure can't draw. In the mid-1980s (1987, by the league's account) Walsh began bringing minority coaches into the 49ers' training camp, to give them NFL experience and a reference from an NFL staff at a time when the profession was almost entirely white. The league later adopted the program for every team; it now carries his name as the Bill Walsh NFL Diversity Coaching Fellowship. Dennis Green and Ray Rhodes, two of the first Black head coaches of the modern era, got their NFL starts on Walsh's staffs. Herm Edwards was the first graduate of the program to become an NFL head coach, and Marvin Lewis, Lovie Smith, Mike Tomlin and Raheem Morris followed.²⁶ That might be the most important tree he planted.

What survives

Forty-odd years later, almost nothing about the 1981 49ers looks modern: the split backs, the quarterback under center on nearly every snap, the seven-step drop. And almost everything they *did* is still here.

- **The short pass as a run.** Every modern offense has a set of throws that does the running game's job: the quick game, the screen, and the RPO, which goes one step further and lets the quarterback choose between the handoff and the throw after the snap (RPOs). The fingerprint chart above is the league running Walsh's offense.
- **Backs as receivers.** Roger Craig's 1,000/1,000 season of 1985 was matched in 1999 by Marshall Faulk and in 2019 by Christian McCaffrey, then with Carolina, who was the 2025 Comeback Player of the Year in Kyle Shanahan's descendant of the same offense in San Francisco.²⁷
- **Timing and placement.** Quarterbacks are still taught to throw on rhythm, with their feet as the clock, and to put the ball where the receiver can keep running. The Quarterback Play chapter is built on those ideas.
- **The script and the rehearsal.** Nearly every NFL play-caller scripts the opening plays, and every staff plans its practice week to the minute. Andy Reid's index card of first-and-second-down calls, in *The Game Plan*, is Walsh's script, one branch down.
- **The language.** The long, explicit West Coast call ("Spider 2 Y Banana") lives on in the Shanahan, McVay and Reid families, though many staffs now shorten it with wristbands and one-word calls (Offensive Systems).

What faded is mostly the *look*. The two-back pro set gave way to one back and three receivers, and then to the shotgun; the fullback nearly disappeared (Shanahan's 49ers are an exception). But the look is coming partly back: in 2025, offenses lined up with fewer than three wide receivers on 41.7% of plays, the most since the NFL's tracking data began in 2016, and the quarterback was under center on about a third of snaps, up from about 27% in 2023.²⁸ The descendants of Walsh's tree are leading that return, for reasons the Shanahan case study explains.

²⁶San Francisco 49ers, "A look into the history and diversity of Bill Walsh's coaching tree": <https://49ers.com/news/a-look-into-the-history-and-diversity-of-bill-walsh-s-coaching-tree> (dates the formal fellowship to 1986, begun with Dr. Harry Edwards and Rodney Knox; Walsh hiring Dennis Green in 1979 and giving Ray Rhodes his first NFL job; every team now participates); San Francisco 49ers, "NFL re-names program after Walsh" (January 28, 2009): <https://www.49ers.com/news/nfl-re-names-program-after-walsh-491253> (the idea "introduced to the league by Walsh in 1987"; Herm Edwards, Marvin Lewis, Lovie Smith, Mike Tomlin and Raheem Morris as graduates who became head coaches). Edwards as the first graduate named an NFL head coach (Jets, 2001): New England Patriots (NFL release), "Bengals' Lewis joins Jets' Edwards as head coach grads of program": <https://www.patriots.com/news/nfl-minority-coaching-fellowship-attracts-81-participants-170256>.

²⁷"Christian McCaffrey," Wikipedia: https://en.wikipedia.org/wiki/Christian_McCaffrey (2019 with Carolina: 1,387 rushing and 1,005 receiving yards, the third player with 1,000 of each in one season after Roger Craig in 1985 and Marshall Faulk in 1999); 2025 Comeback Player of the Year with San Francisco: course fact sheet ([planning/FACTS-current.md](#), §2).

²⁸NFL Next Gen Stats via NFL.com, "NFL trend watch: Heavy personnel could shape Super Bowl LX showdown between Patriots and Seahawks": <https://www.nfl.com/news/nfl-trend-watch-heavy-personnel-could-shape-super-bowl-lx-showdown-between-patriots-and-seahawks> (fewer than three wide receivers on 41.7% of plays in 2025, the most since 2016); under-center share from FTN charting ([qb_location == "U"](#)): 27.3% in 2023, 33.6% in 2025, course calculation in [planning/FACTS-current.md](#), §10.

Watch for it on Sunday

💡 Watch for it: a West Coast census

Pick a team from the tree (the 49ers, Rams, Packers, Chiefs, Vikings, Bengals, Jaguars or any of the others in the 2026 box) and chart its first-half first and second downs. For every pass, note two things:

- **Where was the ball caught, or where would it have been?** Behind the line, one to five yards past it, or deeper? Count the first two together as *short horizontal throws*.
- **Was the receiver running when he caught it?** If yes, how far did he go after the catch?

Then do the arithmetic. Across the league in 2025, 56% of early-down throws in competitive games travelled five yards or less in the air; teams ranged from 46% (CHI) to 64% (PIT). A team well above the league figure is leaning on the horizontal game; a team well below it is throwing downfield on early downs and running to set up its short game.

Post-snap, on the replay: pick one short completion and watch the receiver's hands. Did the ball arrive in front of him, at chest height, so he never slowed down (A in Figure 4), or behind him? And look for the backs: count how many times a running back releases into a route after checking for a blitz. That's the 1984 49ers, forty years on.

Predict the play

💡 Predict the play 1: eight in the box

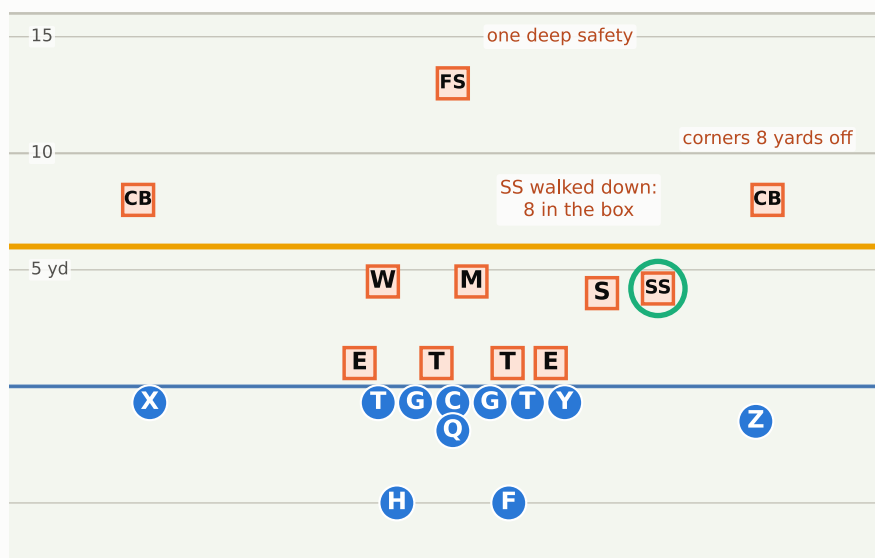


Figure 8: Predict the play 1. 1984, second-and-6 near midfield. The 49ers are in split backs. The defense has walked its strong safety (SS, ringed) down beside the linebackers, so eight defenders stand within about five yards of the line, with one deep safety and both cornerbacks 7 to 8 yards off the receivers.

It's 1984 and you're calling plays for Walsh. Second-and-6 near midfield, split backs. The defense has brought a safety down: eight men near the line, one deep, and the cornerbacks playing well off your receivers. A 1970s offense runs here. What does the West Coast offense do, and where does the ball go?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: press man against the backs

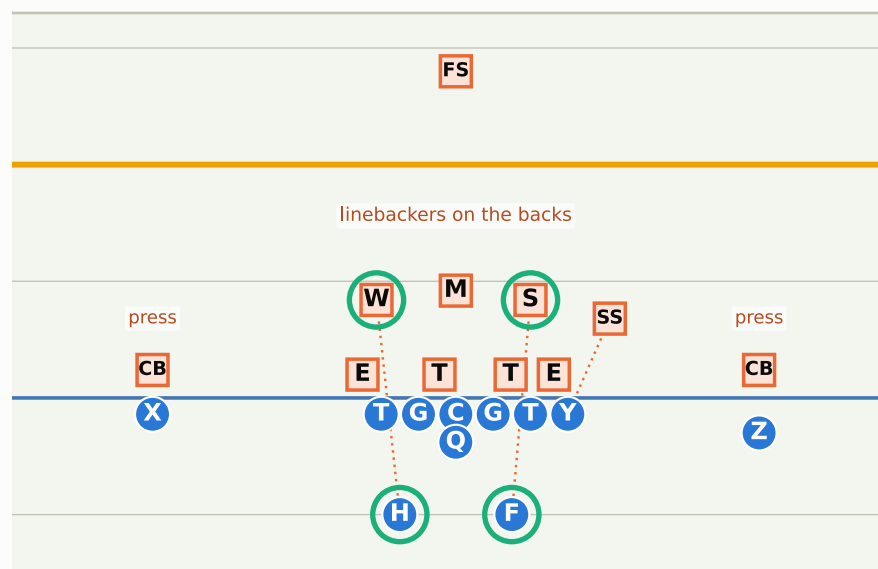


Figure 9: Predict the play 2. 1986: a physical defense in man coverage. Both cornerbacks press X and Z at the line, the strong safety is on the tight end, and two linebackers (W and S, ringed) are assigned to the two backs. One free safety stays deep. The halfback (H) and fullback (F) are ringed.

The Giants of the mid-1980s did something like this to the 49ers: press the wide receivers so the timing breaks down, and hit everybody. You're Walsh. Your receivers will be jammed at the line, and the pressure will come fast. Which of your players is in the best matchup, and what kind of route do you give him?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: the first option is gone

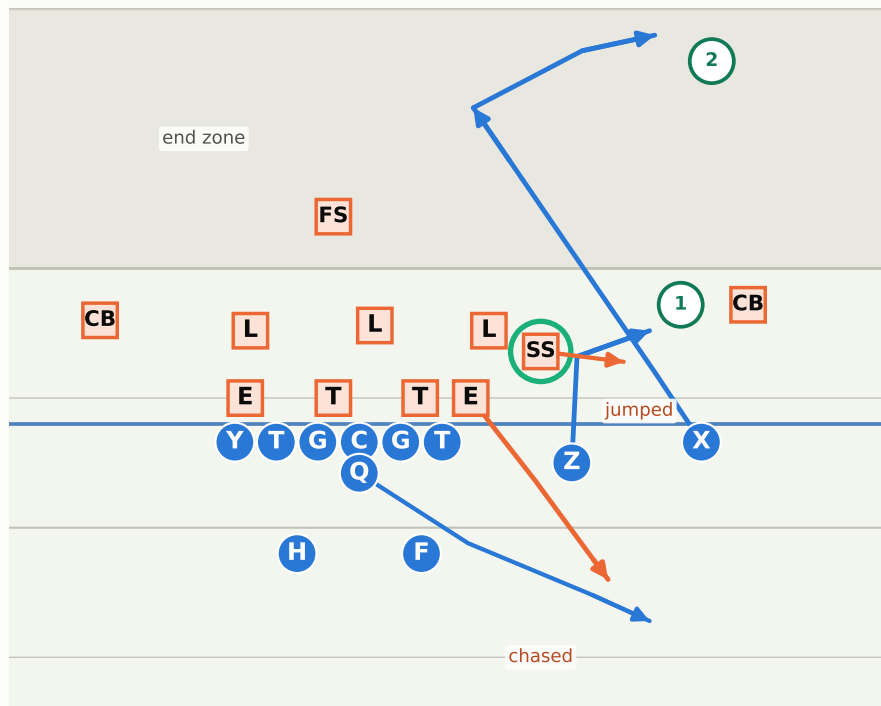


Figure 10: Predict the play 3. Third-and-goal at the 6 in the final minute, trailing by 6. The offense calls a sprint-out right: the slot receiver Z runs an out as the first option (1), and the wide receiver X runs across the end zone and back along the end line as the second (2). The ringed defender has jumped the out and the quarterback is being chased toward the sideline.

You know this play now. But imagine it on a different day: the first option didn't slip, the defender simply jumped the out, and the quarterback is running out of room. It's third down, you're trailing by 6, and there are about fifty seconds left. Where should the ball go, and what's the worst thing he can do?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Walsh's problem was a team that couldn't run the ball, and his answer was to make the pass do the run's job.** The horizontal passing game uses short, timed throws to the edges and across the field, caught by receivers who are already running, so the ball beats the defense's pursuit and the tackle is made in space by a smaller man.
- **Timing, yards after the catch and ball placement were design goals, not happy accidents.** Drops matched route depths and the quarterback's feet ran his progression; routes caught receivers moving away from defenders; and the ball was thrown to the spot that let them keep running. The bill was an offense that needed an accurate quarterback and perfect rehearsal.
- **Scripting and Walsh's practice methods were the system's insurance.** A timing offense can't improvise, so he planned the week, the calls and the footwork in advance, and rehearsed even the backup options. The Catch was Sprint Right Option's *second* option, practised all season.
- **The data still backs him.** On early downs from 2023 to 2025, a pass caught one to five yards past the line succeeded more often than a designed run, and a screen did about the same job as a run.
- **Defenses answered with press and physical play, the zone blitz, rally-and-tackle zones like Tampa 2, and by learning the language,** and today's two-high shells are the latest round.
- **The coaching tree is usually called the largest in football:** ten of the 32 head coaches in 2025 and twelve in 2026 came from its branches. What travels down it is a way of working more than a playbook.

Connections

Builds on: 1978 and the Passing Revolution (the rules that made the timing game possible), Dropback Concepts (drive, curl-flat and how a concept is read), Passing Fundamentals and The Quick Game (timing, rhythm and the Bengals origin story), and Offensive Systems (the West Coast language).

Sets up: the rest of Part 12, starting with The Shanahan Tree (wide zone, boots and YAC over expected, Walsh's offense by way of Denver) and Andy Reid and the Kansas City Chiefs (the Holmgren branch, gone spread); the Belichick case study, Game-Plan Football, is the other great system of the era, built on the opposite philosophy. Appendix E maps the coaching trees in full.

Go deeper

- Bill Walsh with Glenn Dickey, *Building a Champion* (1990): Walsh's own account of rebuilding the 49ers, including his practice planning.
- Bill Walsh with Brian Billick and James A. Peterson, *Finding the Winning Edge* (1997): the coaches' manual: scripting, practice, the passing game and how a staff runs a season.
- Bill Walsh with Steve Jamison and Craig Walsh, *The Score Takes Care of Itself* (2009): his leadership philosophy, built around what he called a "Standard of Performance."
- Tim Layden, *Blood, Sweat and Chalk* (2010): the chapter on the West Coast offense, with the coaches in their own words.
- David Harris, *The Genius: How Bill Walsh Reinvented Football and Created an NFL Dynasty* (2008): the full biography.

Answers to Predict the play

Answer 1.

It throws, short and outside, and treats the throw as the run. Count the box: eight defenders against seven blockers (five linemen, the tight end and a back) means a run is blocked one man short, and the defense built this alignment to stop it. But the same alignment leaves the edges thin. With the corners 8 yards off and only one deep safety, a quick throw outside, a hitch or slant to X, or the flat to a back, gains five or six yards before anyone can tackle, and the eight players in the box are all on the wrong side of the line of scrimmage to chase it. Second-and-6 only needs the six.

This is the core of the chapter in one decision: the West Coast offense doesn't argue with eight in the box, it throws the ball to where the box isn't. The defense's counter is to keep eight in the box *and* press the corners, which leaves no deep help at all; that's when Walsh went over the top.

Answer 2.

The backs. Nobody can press a running back, because he starts five yards behind the line. And the men assigned to them are linebackers, who are bigger and slower than a back like Roger Craig. So the answer is to throw to the backs on routes that use that edge: a flat or swing that makes the linebacker run to the sideline, or an angle route that starts toward the flat and snaps back inside, away from the linebacker's momentum. Against man coverage, crossing routes and rubs also help receivers escape their defenders, and a sprint-out moves the launch point away from the rush.

The trade-off is protection. If the backs run routes, they aren't blocking, and a defense playing press man often blitzes. That's what the check-release is for: each back looks for his blitzer first and releases only if nobody comes. Against a blitz the quarterback has to throw before the unblocked rusher arrives, which is why the quick throw to the back is the first thing he looks at.

Answer 3.

To the second option, high and to the back of the end zone, or nowhere. The receiver coming back along the end line is the play's built-in answer to exactly this: the defender who jumped the out has left the back of the end zone to one man covering a receiver who is moving away from him. The throw goes high, where only the taller receiver can catch it, and if he can't, it sails out of bounds.

The worst outcomes are a sack and an interception, and for different reasons. A sack loses yards and, with no timeouts to spare, maybe the clock; an interception ends the game. An incompleteness is fine: it brings up fourth-and-goal from the 6 with time for one more play. That's why the throw at the end of The Catch was placed so that nothing bad could happen. The quarterback's job on a broken play is to keep the next play alive while giving the designed answer a chance.